

Optimal Control

LQR Optimal Control Task 2M

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Var 6

6	$\begin{bmatrix} 2 & 3 \\ -7 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 8 \end{bmatrix}$	$\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$
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System Model

$$A = \begin{bmatrix} 2 & 3 \\ -7 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$B_f = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$Q = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$$

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Use

$$U = K * x$$

disturbance input

$$f(t) = 10\sin(6t) + 5\cos(2t) + 4\cos(3t) + 3\cos(8t)$$

Initial condition

$$X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

H- ∞ Control Design

controller is designed using the Riccati equation

$$A^T P + P A + Q - P B B^T P + \gamma^{-2} P B_f B_f^T P = 0$$

Compute gamma min almost = 7.645

So

Let gamma = 8

positive definite solution of the Riccati equation is

$$P = \begin{bmatrix} 7.5334 & -12.8020 \\ -12.8020 & 35.0569 \end{bmatrix}.$$

feedback gain matrix is obtained from

$$K = -B^T P$$

$$K = [-7.5334; 12.8020]$$

$$u(t) = -7.5334x_1(t) + 12.8020x_2(t)$$

Set simulink model

参数

A:

B:

C:

D:

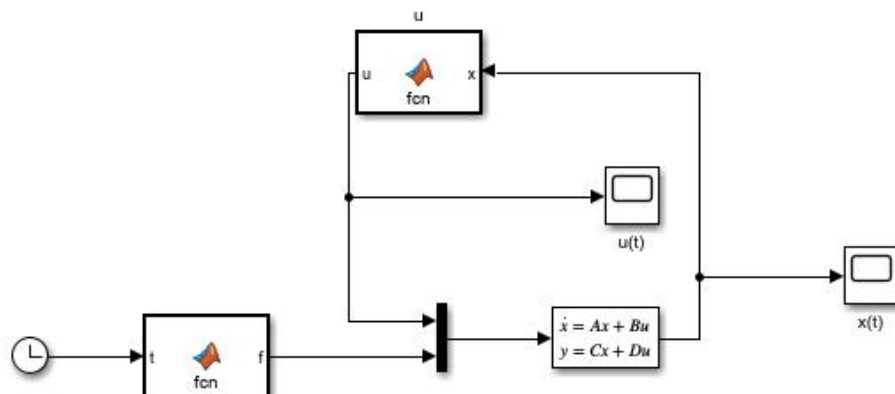
初始条件:

参数可调性: 自动

☐ 允许最初指定为零的 D 矩阵具有非零值

绝对容差:

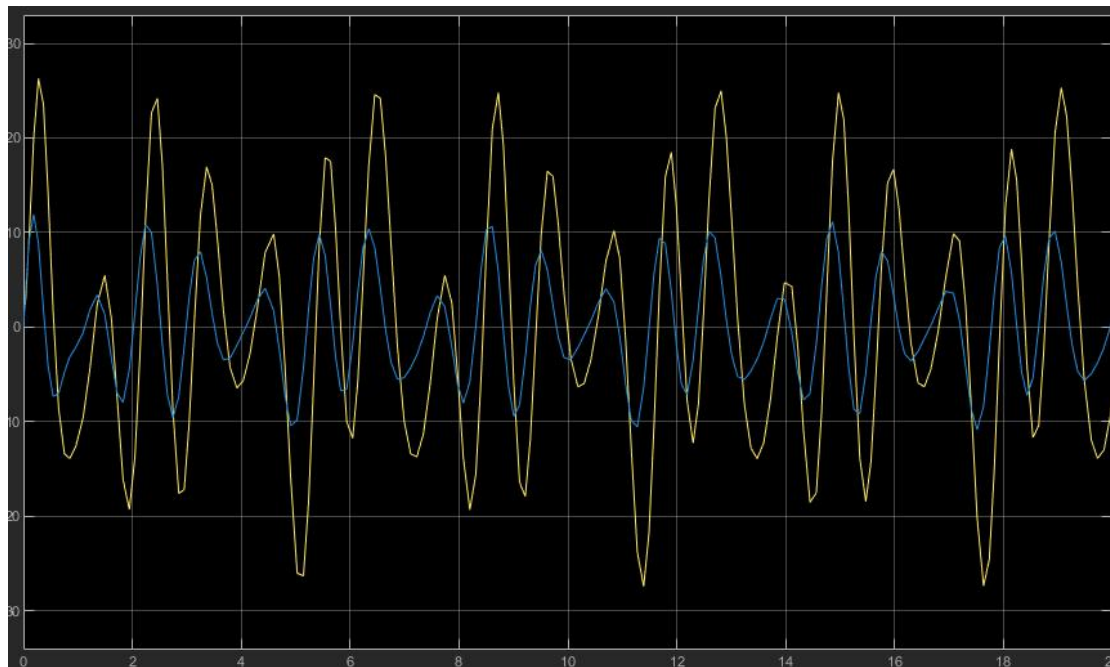
状态名称: (例如, 'position')



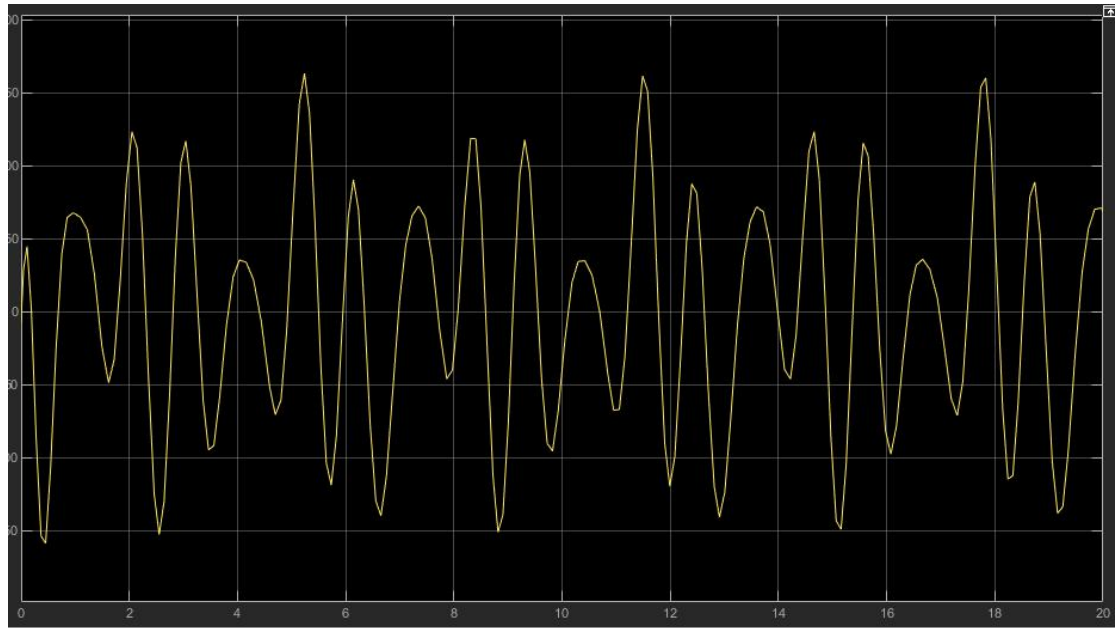
```
Op1b6 ▶ u
1 function u = fcn(x)
2   K = [-7.5334 12.8020];
3   u = K*x;
4
```

```
Op1b6 ▶ MATLAB Function1
1 function f = fcn(t)
2
3   f = 10*sin(6*t) ...
4       + 5*cos(2*t) ...
5       + 4*cos(3*t) ...
6       + 3*cos(8*t);
```

Input



Out put



Conclusion:

In this experiment, an H_∞ state feedback controller was designed and implemented in Simulink. The simulation results show that the controller can stabilize the system and reduce the effect of disturbances.

Therefore, the designed controller is effective and the closed-loop system has good robustness.